Innovation Cell

NIT Raipur

Software Requirements Specification

I Cell Inventory Management System

Document Version	0.1 (Draft)
Document Status	Requirements Baseline – Draft
Prepared For	Innovation Cell, NIT Raipur
Prepared By	Aditya Sahu, Core Coordinator, Technical Domain
Date	October 8, 2026

This document defines the functional, non-functional, business, and data requirements for the proposed I Cell Inventory Management System.

Contents

1 Introduction **3**
1.1 Purpose . . . 3
1.2 Scope . . . 3
1.3 Intended Audience . . . 3
1.4 Definitions and Abbreviations . . . 4

2 System Overview **4**
2.1 Problem Statement . . . 4
2.2 System Objectives . . . 5
2.3 High-Level System Workflow . . . 5
2.4 Assumptions . . . 6
2.5 Out of Scope for Initial Version . . . 6

3 User Roles and Access Control **7**
3.1 Roles . . . 7
3.2 Permission Matrix . . . 7

4 Functional Requirements **8**
4.1 Authentication and User Registration . . . 8
4.2 User Management . . . 8
4.3 Inventory Management . . . 9
4.4 Inventory Visibility . . . 10
4.5 Category Management . . . 11
4.6 Equipment Requests . . . 11
4.7 Request Approval and Rejection . . . 12
4.8 Equipment Issue . . . 12
4.9 Equipment Return . . . 13
4.10 Damage and Loss Management . . . 13
4.11 History and Audit Logging . . . 14
4.12 Dashboard and Reporting . . . 15

5 Non-Functional Requirements **15**
5.1 Security . . . 15
5.2 Data Integrity and Consistency . . . 16
5.3 Performance . . . 16
5.4 Availability and Reliability . . . 16
5.5 Backup and Recovery . . . 17
5.6 Usability . . . 17
5.7 Maintainability . . . 17
5.8 Scalability . . . 18

6 Business Rules **18**

7 Data Requirements **19**
7.1 Core Entities 19
7.2 Item Data Model 19
7.3 Request Data 20
7.4 Issue and Return Data 20
7.5 Liability Data 21

8 External Interface Requirements **21**
8.1 User Interface 21
8.2 Email Interface 22

9 Security and Privacy Considerations **22**

10 Constraints **22**

11 Future Scope **23**

12 Open Items for Requirements Baseline **23**

13 Revision History **23**

1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for a web-based Inventory Management System for Innovation Cell (I Cell), NIT Raipur.

The system is intended to provide a centralized mechanism for maintaining information about reusable hardware components and equipment owned by I Cell, managing equipment requests, obtaining approvals, recording issue and return transactions, tracking damaged or lost equipment, and maintaining an auditable history of inventory-related activities.

This document is intended to serve as a common reference for the Core Coordinators, the student development team, the Overall Coordinators, and Faculty Advisors during design, implementation, testing, deployment, and future maintenance.

1.2 Scope

The proposed system will manage the lifecycle of reusable I Cell equipment from inventory registration through member request, approval, issue, return, and, where applicable, damage/loss and liability resolution.

The system will support multiple categories and physical storage locations. It will use role-based access control to ensure that users can perform only the operations authorized for their role.

The initial system will not include purchase/vendor management or QR/barcode-based asset identification.

1.3 Intended Audience

This document is intended for:

- Faculty Advisors of I Cell.
- Overall Coordinators.
- Head Coordinators and Core Coordinators.
- Student developers responsible for implementing the system.
- Future I Cell technical teams responsible for maintenance and extension of the system.

1.4 Definitions and Abbreviations

Term	Definition
I Cell	Innovation Cell, NIT Raipur.
SRS	Software Requirements Specification.
Member	An authenticated I Cell user who can request equipment but does not have administrative inventory permissions.
Core Coordinator	A 3rd-year I Cell coordinator.
Head Coordinator	A 4th-year I Cell coordinator who is not an Overall Coordinator and does not have inventory administrative privileges.
Overall Coordinator (OC)	A 4th-year I Cell Overall Coordinator authorized to approve requests, process issues/returns, and manage inventory.
Faculty Advisor	Faculty member/advisor authorized to perform administrative inventory operations.
Item	A type of reusable equipment/component tracked by the system.
Request	A member's request for one or more equipment types and quantities.
Issue	A record indicating that approved equipment has been physically provided to a member.
Return	A record indicating the return of equipment previously issued to a member.
Liability	A record of a member's responsibility for damaged or lost equipment and its eventual resolution.
Audit Log	An immutable or appropriately protected record of significant system actions performed by users.

2 System Overview

2.1 Problem Statement

I Cell maintains reusable hardware components and other equipment that may be used by its members for projects, events, workshops, and other activities. A centralized system is required to maintain an accurate inventory and provide accountability for equipment issued to members.

The system must answer, among other questions:

- What equipment does I Cell own?
- How much of each equipment type is available?

- Where is inventory physically stored?
- Which members currently hold equipment?
- Which requests are awaiting approval?
- What equipment has been returned, damaged, or lost?
- Who approved, issued, returned, or modified an inventory record?
- Which liabilities are still unresolved?

2.2 System Objectives

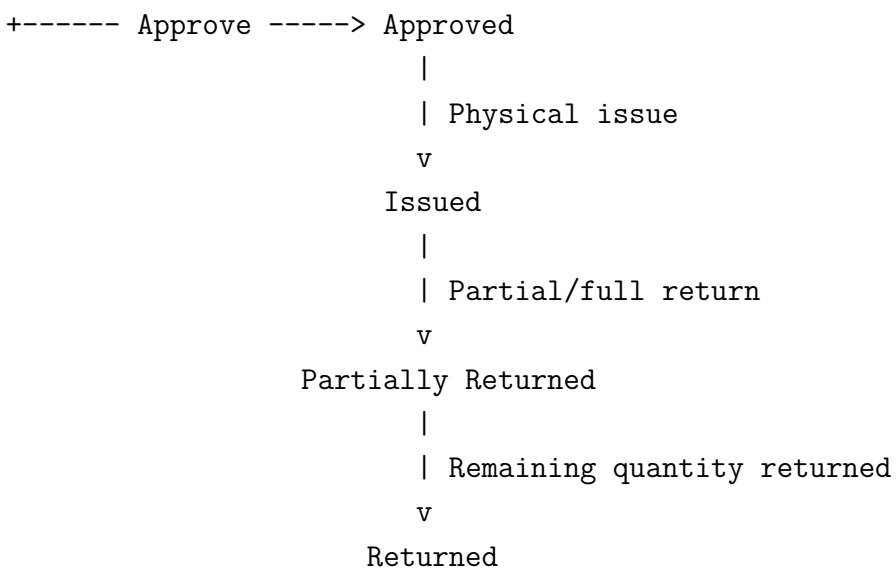
The primary objectives are:

1. Centralize I Cell inventory information.
2. Simplify and formalize the equipment request and approval process.
3. Maintain accurate records of issued and available quantities.
4. Track partial approvals and partial returns.
5. Maintain accountability for damaged or lost equipment.
6. Preserve historical inventory and transaction information.
7. Enforce role-based authorization.
8. Provide an audit trail for significant system actions.
9. Provide a maintainable system that can be handed over to future I Cell technical teams.

2.3 High-Level System Workflow

The principal equipment lifecycle is:

```
Member
  |
  | Create request
  v
Pending Request
  |
  +------ Cancel ------> Cancelled
  |
  +------ Reject ------> Rejected
  |
```



Damage and loss are handled as separate outcomes of the return/loss process and may create a liability for the responsible member.

2.4 Assumptions

- All users of the system are associated with I Cell.
- Users authenticate using their NIT Raipur institutional email address.
- Inventory items are reusable assets; the system does not initially model consumable stock.
- Equipment is tracked by quantity rather than by individual physical-unit serial numbers.
- A request can contain multiple equipment types.
- Approved quantities can differ from requested quantities.
- A member can have multiple simultaneous active issues.
- There is no mandatory return deadline in the initial system.

2.5 Out of Scope for Initial Version

The following are explicitly outside the initial scope unless subsequently approved as requirements:

- Purchase order and vendor management.
- QR/barcode-based identification.
- Individual serial-number tracking of physically identical units.

- Online payment processing for liability settlement.
- Automatic WhatsApp notifications.

3 User Roles and Access Control

3.1 Roles

The system shall recognize the following roles:

1. Member (2nd-year and other ordinary I Cell members).
2. Core Coordinator (3rd year).
3. Head Coordinator (4th year, not an Overall Coordinator).
4. Overall Coordinator (4th year).
5. Faculty Advisor.

Academic year and application role shall be treated as separate concepts. Permissions shall be determined by the assigned application role rather than by academic year alone.

3.2 Permission Matrix

Capability	Member	Core	Head	OC	Faculty
Browse inventory	✓	✓	✓	✓	✓
Create request	✓	✓	✓	✓	✓
Cancel own pending request	✓	✓	✓	✓	✓
View own history	✓	✓	✓	✓	✓
View other members' history	–	–	–	✓	✓
Approve/reject requests	–	–	–	✓	✓
Process equipment issue	–	–	–	✓	✓
Process equipment return	–	–	–	✓	✓
Record damage/loss	–	–	–	✓	✓
Resolve liability	–	–	–	✓	✓
Add/edit inventory	–	–	–	✓	✓
Delete inventory	–	–	–	✓	✓
Add/edit categories	–	–	–	✓	✓
Manage users	–	–	–	✓	✓
View audit logs	–	–	–	✓	✓

Authorization shall be enforced by the backend and shall not rely solely on frontend visibility controls.

4 Functional Requirements

4.1 Authentication and User Registration

FR-AUTH-01
The system shall allow users to authenticate using their NIT Raipur institutional email address and password.

FR-AUTH-02
The system shall restrict registration to valid NIT Raipur institutional email addresses.

FR-AUTH-03
A prospective member shall be able to submit a registration request using their institutional email address.

FR-AUTH-04
A registration request shall remain pending until approved by an authorized administrator.

FR-AUTH-05
Upon approval, the system shall create/activate the user's account and provide an initial credential through the registered institutional email.

FR-AUTH-06
The initial credential shall be temporary and the user should be required to establish a new password on first login.

FR-AUTH-07
Passwords shall be stored only as secure password hashes and shall never be stored in plaintext.

FR-AUTH-08
The system shall support logout and secure session termination.

FR-AUTH-09
An administrator shall be able to deactivate a user without deleting the user's historical transaction records.

4.2 User Management

FR-USER-01
The system shall maintain user identity, institutional email, academic year, role, and account status.

FR-USER-02
Authorized administrators shall be able to approve pending registrations.

FR-USER-03
Authorized administrators shall be able to assign or modify a user's application role.

FR-USER-04
Authorized administrators shall be able to deactivate users.

FR-USER-05
Deactivated users shall not be able to initiate new requests or perform other authenticated operations.

FR-USER-06
Historical records associated with a deactivated user shall remain accessible to authorized users.

4.3 Inventory Management

FR-INV-01
The system shall maintain an inventory record for each equipment type.

FR-INV-02
An inventory item shall contain, at minimum:

- unique ID,
- name,
- category,
- description,
- total quantity,
- issued quantity,
- and timestamps for creation and modification.

FR-INV-03
Available quantity shall be represented as the quantity currently available for issue.

FR-INV-04
The system shall maintain inventory distributed across multiple physical locations.

FR-INV-05
The system shall maintain the quantity of an item at each applicable location.

FR-INV-06
Authorized administrators shall be able to add inventory items.

FR-INV-07
Authorized administrators shall be able to edit inventory item details.

FR-INV-08
Authorized administrators shall be able to adjust inventory quantities, with the adjustment recorded in the audit/history mechanism.

FR-INV-09
Authorized administrators shall be able to change the physical location of inventory or its location-wise quantity distribution.

FR-INV-10
Inventory deletion shall be restricted to authorized administrators.

FR-INV-11
The system shall preserve historical transaction integrity when an item with existing historical records can no longer be actively used. Where hard deletion would compromise historical records, the implementation shall use an appropriate inactive/retired representation.

4.4 Inventory Visibility

FR-VIS-01
Members, Core Coordinators, and Head Coordinators shall be able to view the following inventory information:

- item ID,
- name,
- category,
- description,
- total quantity,
- issued quantity,
- available quantity.

FR-VIS-02
Members, Core Coordinators, and Head Coordinators shall not be shown internal inventory-management information such as storage location or management timestamps through normal inventory views.

FR-VIS-03
Overall Coordinators and Faculty Advisors shall be able to access additional inventory-management information required for their duties.

FR-VIS-04
Backend responses shall enforce the same information-access restrictions as the user interface.

4.5 Category Management

FR-CAT-01
The system shall organize inventory items into categories.

FR-CAT-02
Overall Coordinators and Faculty Advisors shall be able to create categories.

FR-CAT-03
Overall Coordinators and Faculty Advisors shall be able to edit categories.

FR-CAT-04
Category deletion shall not be permitted if it would invalidate existing inventory records; an inactive representation shall be used where appropriate.

4.6 Equipment Requests

FR-REQ-01
An authenticated I Cell member shall be able to create an equipment request.

FR-REQ-02
A request shall support one or more equipment types.

FR-REQ-03
Each requested equipment type shall have an independently specified requested quantity.

FR-REQ-04
A member shall be allowed to have multiple active requests.

FR-REQ-05
A member shall be allowed to request equipment even when the same equipment type is already issued to them.

FR-REQ-06
A member shall be able to view their own request history.

FR-REQ-07
A member shall be able to cancel a request while it is pending.

FR-REQ-08
A pending request shall not be edited after submission. A member who wishes to change it shall cancel the existing request and create a new request.

FR-REQ-09
A request shall record its requester, creation time, current state, and associated requested

items.

FR-REQ-10
The system shall prevent an issue from exceeding the quantity available under the applicable inventory rules.

4.7 Request Approval and Rejection

FR-APR-01
Only Overall Coordinators and Faculty Advisors shall be authorized to approve or reject requests.

FR-APR-02
Either an Overall Coordinator or a Faculty Advisor shall be able to independently approve or reject a request.

FR-APR-03
Approval shall be performed independently for each requested equipment type and quantity.

FR-APR-04
The approved quantity for an item may be less than, equal to, or zero relative to the requested quantity.

FR-APR-05
A rejection reason shall be mandatory whenever a requested item is rejected.

FR-APR-06
The system shall record the approving/rejecting user and timestamp.

FR-APR-07
The system shall prevent approval of quantities that cannot be issued under the current inventory state.

FR-APR-08
Approval shall not by itself imply that the equipment has physically been handed over to the member. Physical issue shall be recorded separately.

4.8 Equipment Issue

FR-ISS-01
Only Overall Coordinators and Faculty Advisors shall be authorized to process equipment issues.

FR-ISS-02
An approved quantity shall become issued only when an authorized user records the physical issue.

FR-ISS-03
The issue record shall identify the member, equipment, quantity, issuing user, and issue timestamp.

FR-ISS-04
The system shall increase issued quantity and decrease available quantity when an issue is successfully recorded.

FR-ISS-05
The system shall support multiple active issue records for the same member.

FR-ISS-06
The system shall support multiple active issue records for the same equipment type.

4.9 Equipment Return

FR-RET-01
Only Overall Coordinators and Faculty Advisors shall be authorized to process returns.

FR-RET-02
A return shall be associated with an existing issue.

FR-RET-03
Partial returns shall be supported.

FR-RET-04
The system shall track issued quantity, returned quantity, and outstanding quantity for an issue.

FR-RET-05
A successfully returned item in usable condition shall increase available inventory and reduce the outstanding issued quantity.

FR-RET-06
The return record shall identify the processing user and timestamp.

FR-RET-07
There shall be no mandatory return deadline in the initial version.

4.10 Damage and Loss Management

FR-DL-01
The system shall support recording equipment returned in damaged condition.

FR-DL-02
The system shall support recording equipment that is lost.

FR-DL-03
Once damage or loss is confirmed, the affected quantity shall no longer be considered

usable available inventory.

FR-DL-04
Damage/loss events shall remain represented in historical records and shall not be silently removed by inventory quantity changes.

FR-DL-05
The system shall create a liability record for the responsible member when applicable.

FR-DL-06
Liability shall have at least the states *Pending* and *Resolved*.

FR-DL-07
A resolved liability shall record the resolution type, including at least *Paid* or *Replaced*.

FR-DL-08
Only Overall Coordinators and Faculty Advisors shall be authorized to record and resolve damage/loss liabilities.

FR-DL-09
The system shall not perform online payment processing as part of the initial liability workflow.

4.11 History and Audit Logging

FR-AUD-01
The system shall maintain an audit history of significant system actions.

FR-AUD-02
Audit records shall identify the actor, action, affected entity, affected entity identifier where applicable, and timestamp.

FR-AUD-03
Important changes shall retain sufficient information to determine the relevant previous and new state where applicable.

FR-AUD-04
Audit records shall include, at minimum, important authentication, user-management, request, approval, inventory, issue, return, damage, loss, and liability actions.

FR-AUD-05
Members shall be able to view their own request and equipment history but shall not be able to access another member's private transaction history.

FR-AUD-06
Overall Coordinators and Faculty Advisors shall be able to access the history necessary for inventory administration and accountability.

4.12 Dashboard and Reporting

FR-DASH-01
The system should provide an administrative dashboard showing useful inventory and workflow statistics.

FR-DASH-02
The dashboard should include, where applicable:

- total inventory,
- available quantity,
- currently issued quantity,
- pending requests,
- damaged/lost quantities,
- pending liabilities.

FR-DASH-03
The system shall not treat equipment as overdue in the initial version because a mandatory return deadline has not been defined.

5 Non-Functional Requirements

5.1 Security

NFR-SEC-01
The system shall enforce authentication for protected operations.

NFR-SEC-02
The system shall enforce role-based authorization on the server side.

NFR-SEC-03
Passwords shall be securely hashed using an industry-standard password hashing mechanism.

NFR-SEC-04
The system shall validate and sanitize user-controlled input.

NFR-SEC-05
Protected operations shall not be made accessible merely by manipulating frontend requests.

NFR-SEC-06
Sensitive authentication and application traffic shall use secure transport in production.

NFR-SEC-07
The system shall protect audit records from unauthorized modification or deletion.

5.2 Data Integrity and Consistency

NFR-DAT-01
Inventory quantities shall remain internally consistent.

NFR-DAT-02
The system shall prevent issuing quantities that exceed available inventory.

NFR-DAT-03
Inventory-changing operations shall be atomic so that partial database updates do not leave inconsistent state.

NFR-DAT-04
Historical transaction records shall not be silently altered merely to reflect current inventory state.

NFR-DAT-05
Concurrent operations affecting the same inventory shall be handled so that overselling/over-issuing of available quantity cannot occur.

5.3 Performance

NFR-PERF-01
Common operations such as inventory browsing, search, request creation, approval, issue, and return should provide responsive interaction under the expected I Cell workload.

NFR-PERF-02
Database queries shall be designed to remain efficient as transaction history grows over multiple years.

NFR-PERF-03
The system shall support multiple concurrent users without compromising data integrity.

5.4 Availability and Reliability

NFR-REL-01
The application should be available during normal I Cell operational periods.

NFR-REL-02
The system shall handle application or database errors without silently losing completed inventory transactions.

NFR-REL-03
Critical inventory operations shall fail safely when they cannot be completed.

5.5 Backup and Recovery

NFR-BACK-01
Inventory, user, request, issue, return, liability, and audit data shall be included in an appropriate backup strategy.

NFR-BACK-02
The project shall document a procedure for restoring the application database from a valid backup.

NFR-BACK-03
Backup and recovery procedures shall be tested periodically.

5.6 Usability

NFR-USE-01
The system shall provide a clear and consistent interface for members and administrative users.

NFR-USE-02
Common workflows such as browsing equipment, submitting requests, approving requests, issuing equipment, and recording returns should require a reasonable number of user interactions.

NFR-USE-03
The system should be usable on common desktop and mobile screen sizes.

5.7 Maintainability

NFR-MNT-01
The codebase shall use a modular architecture with clear separation of responsibilities.

NFR-MNT-02
The project shall maintain documentation sufficient for handover to a future I Cell technical team.

NFR-MNT-03
The project shall use version control.

NFR-MNT-04
The project shall define consistent coding, naming, and API conventions.

NFR-MNT-05
Critical business logic shall be covered by automated tests where practical.

5.8 Scalability

NFR-SCL-01
The system shall support growth in the number of inventory items, users, and historical transactions without requiring fundamental redesign for the expected scale of I Cell.

NFR-SCL-02
The design should permit future features such as QR/barcode support, notifications, or richer reporting without requiring replacement of the core inventory model.

6 Business Rules

Rule ID	Business Rule
BR-01	Only authorized I Cell users may access protected system functions.
BR-02	Members, Core Coordinators, and Head Coordinators may request equipment but may not approve, issue, return, or manage inventory.
BR-03	Overall Coordinators and Faculty Advisors may independently approve/reject requests and process issue/return operations.
BR-04	A request may contain multiple equipment types.
BR-05	Approval quantity is determined independently for each requested equipment type.
BR-06	A member may have multiple active requests and active issues.
BR-07	A member may request equipment already currently issued to them.
BR-08	A pending request can be cancelled but cannot be edited; changes require cancellation and creation of a new request.
BR-09	Rejection of a requested item requires a rejection reason.
BR-10	Partial returns are allowed.
BR-11	There is no mandatory return deadline in the initial system.
BR-12	Damaged or lost equipment is the responsibility of the member to the extent defined by I Cell policy.
BR-13	Confirmed damaged/lost quantity is removed from usable inventory.
BR-14	Damage/loss creates a liability record that can remain pending until resolved.
BR-15	Liability may be resolved by payment or replacement, as applicable.
BR-16	Users who leave I Cell are deactivated rather than deleted.

BR-17	Historical records must remain associated with deactivated users.
BR-18	Equipment is tracked by quantity rather than by individual physical unit.
BR-19	Inventory may be distributed across multiple physical locations.
BR-20	Only Overall Coordinators and Faculty Advisors manage categories.
BR-21	Significant inventory and workflow actions must be auditable.

7 Data Requirements

7.1 Core Entities

The initial conceptual data model is expected to contain at least the following entities:

- User
- Category
- Item
- Location / Item Location
- Request
- Request Item
- Issue
- Return
- Damage/Loss Record
- Liability
- Audit Log

7.2 Item Data Model

At the conceptual level, an item contains:

Attribute	Description	Visibility
ID	Unique identifier for the equipment type.	Member
Name	Name of the equipment.	Member
Category	Category to which the equipment belongs.	Member
Description	Description/specification of the equipment.	Member

Total Quantity	Total quantity currently owned/recognized by I Cell.	Member
Issued Quantity	Quantity currently issued to members.	Member
Available Quantity	Quantity currently available for issue.	Member
Location-wise Quantity	Quantity held at each physical location.	Admin
Created At	Record creation timestamp.	Admin
Updated At	Last modification timestamp.	Admin

Available quantity shall be maintained consistently with the underlying inventory and transaction records. The implementation may calculate it rather than storing it redundantly.

7.3 Request Data

A request is expected to contain:

- Request identifier.
- Requester.
- Creation timestamp.
- Request state.
- One or more request-item records.

Each request-item record shall contain at least:

- Equipment/item reference.
- Requested quantity.
- Approved quantity.
- Item-level approval/rejection state.
- Rejection reason where applicable.

7.4 Issue and Return Data

An issue shall record:

- Member receiving equipment.
- Equipment type.
- Issued quantity.

- Issuing authorized user.
- Issue timestamp.
- Reference to the relevant approved request.

A return shall record:

- Reference to the issue.
- Returned quantity.
- Return condition.
- Processing authorized user.
- Return timestamp.
- Remarks where required.

The model shall support multiple return records against a single issue to permit partial returns.

7.5 Liability Data

A liability record shall contain enough information to identify:

- Responsible member.
- Related equipment.
- Quantity.
- Damage/loss type.
- Liability status.
- Creation timestamp.
- Resolution timestamp, if resolved.
- Resolution type, such as paid or replaced.
- Relevant remarks.

8 External Interface Requirements

8.1 User Interface

The system shall provide a web-based user interface.

The interface shall provide role-appropriate functionality and information. Administrative operations shall not be exposed to unauthorized users.

8.2 Email Interface

The system shall be capable of sending email to institutional addresses for at least the account-credential workflow.

The final implementation may extend email usage to notifications such as request approval/rejection depending on later requirements.

9 Security and Privacy Considerations

The system contains information about I Cell membership and equipment responsibility. Access to this information shall therefore follow the principle of least privilege.

In particular:

- Members shall not be able to inspect other members' private equipment histories.
- Administrative inventory information shall be restricted to authorized roles.
- Passwords shall never be exposed through application responses or stored in plaintext.
- Audit information shall be protected from ordinary member access.
- Deactivating a user shall not erase historical accountability.

10 Constraints

The initial implementation is intended for I Cell and should avoid unnecessary enterprise-scale complexity.

The project shall:

- prioritize correctness of inventory and accountability data;
- use a maintainable architecture suitable for student developers;
- remain understandable and transferable to future I Cell batches;
- avoid features that are not supported by an identified I Cell requirement.

No specific programming language, framework, database, or deployment platform is mandated by this SRS. Such decisions shall be documented separately as technical design decisions after the requirements baseline is approved.

11 Future Scope

Potential future extensions include:

- QR/barcode-based equipment identification.
- Automated email reminders.
- Additional notification channels.
- Advanced inventory analytics and reporting.
- Return deadlines and overdue tracking.
- Individual physical-unit/serial-number tracking.
- Integration with other I Cell systems.
- Online liability payment integration.

These features are not requirements for the initial release unless explicitly approved and incorporated into a later version of this SRS.

12 Open Items for Requirements Baseline

The following items should be confirmed before the SRS is marked as a final baseline:

1. Exact NIT Raipur institutional email domain and whether all institutional addresses are eligible.
2. Whether registration approval is performed by all OCs/Faculty Advisors or by a smaller designated administrator group.
3. Exact inventory deletion policy for items with historical transactions.
4. Exact policy for deactivating a user who still has outstanding equipment or unresolved liability.
5. Whether the system will send only account credentials initially or also request/transaction notifications.
6. Exact dashboard and report requirements.

These are deliberately left as requirements questions rather than being silently assumed by the development team.

13 Revision History

Version	Date	Description
0.1	October 8, 2026	Initial requirements draft based on discussions with the I Cell Technical/Web Development team.